

PHYS 304 Final Project:

Inverse Odor Mapping: Estimating Odors Through Inactivity

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In this project, we aim to reproduce and analyze results from a study by Singh *et al.* on odor identification using an estimation-by-elimination method. This method assumes inactive olfactory receptors carry more information than active ones by ruling out the presence of all odorants that bind to them. We simulate both binary and weighted models to evaluate whether the original odor mixture can be reconstructed based solely on receptor activity. In the binary case, we assess decoding accuracy as a function of receptor number and odor complexity, implementing optimizations including vectorized operations and GPU acceleration through PyTorch. In the weighted case, we extend the model to consider odorant concentrations, using linear algebra to estimate mixture composition after eliminating inactive receptor interactions. Our results largely support the findings of the original study, showing that odors with realistic complexity can be decoded with relatively few receptors. However, discrepancies in the weighted case suggest that further refinements, such as thresholding weak receptor responses, may improve accuracy. Overall, our model demonstrates how information from inactive receptors can effectively constrain and decode high-dimensional odor mixtures.

I. INTRODUCTION

Humans and other animals use the olfactory system to capture, process, and understand odor information. In order to perceive the odors, animals have odorant receptors located in the olfactory bulb. Odorants can bind to receptor cells via certain chemical bonding, triggering neurons that are connected to these receptor cells. The information provided by all of the receptors is further processed in the brain. Different types of receptors can bond to different odorants, but the number of odorants in nature is large when compared to the number of different types of receptors in animals. The typical number of receptor types is around 50 for flies, around 300 for humans, and around 1000 for rats, mice, and dogs [1] while the number of odorants in nature is of magnitude 10^4 or higher [1].

In order to solve this problem, each receptor is sensitive to several different odorants, creating an encoding of information. There are different models aiming to decode this receptor response to get the presence of odorants. For example, some models suggest that since the olfactory network has similar activity pattern when similar odors are present, we can classify the activity patterns to similar odors [1]. Singh *et al.* explored a different but very simple approach in their article: estimation by elimination. The main idea for this model is that we are assuming that the inactive receptors carry more information than the active ones, since they will completely eliminate the possibility of existence of the odorants that are bound to them. After using all of the inactive receptors as filters, only a small portion of odorants are

present, converting the problem from under-determined to overdetermined.

Singh *et al.* explored this estimation by elimination method in three steps. Firstly, they focused on the presence of an odorant and ignored the concentration. A randomly generated sparse sensing matrix was used with many randomly generated odor vectors to create the receptor response vector. All values in the odor vectors, the receptor response vectors, and the sparse sensing matrix contained only the presence information (0 for absence or 1 for presence). Secondly, a similar approach was used but the concentration of odors was no longer ignored. Thirdly, they studied how noise will affect the decoder which elimination method is used. Finally, they proposed a neural network that represents how an estimation-by-elimination decoder would work in a typical biological system.

II. PROBLEM STATEMENT/SPECIFIC GOALS

In our project, we plan to reproduce the results by Singh *et al.* Due to limited time, we were only able to reproduce the results of identifying presence and concentration of odorants.

III. METHODS AND NUMERICAL METHODS USED

Before going into the details of our methods, we need to clarify the definition of several parameters that we are using in our simulation. We are simulating an olfactory system with N_R different types of receptors and N_L different odorants. The binding of receptors to odorants is represented by the matrix S , which has dimension N_R by N_L . The probability of an odorant binding to a receptor

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is s , which means that each entry in S has a probability of s to be non-zero. Suppose that there are K odorants present in each of the odorant mixtures, then the odor mixture vector $\mathbf{c} = (c_1, c_2, \dots, c_{N_L})$ has K elements that are non-zero. The response of the receptors is recorded by vector $\mathbf{R} = (R_1, R_2, \dots, R_{N_R})$. For both the binary case (only focusing on the existence of odorants) and the weighted case (focusing on both the existence and the concentration of odorants), the response vector is obtained by matrix multiplication of $\mathbf{R} = S \cdot \mathbf{c}$. Then, we are trying to predict the odorant mixture vector from the response vector $\mathbf{R}$. Our predicted result is marked by $\mathbf{c}'$ in order to distinguish from the actual value $\mathbf{c}$.

A. Binary

For the binary case, we focus on the existence of odorants only. Our mission is to find out under different values of N_R and K , whether the predicted odorant mixture $\mathbf{c}'$ is the same as the input odorant mixture $\mathbf{c}$. The whole process can be split into the encoding and decoding processes.

As shown in Figure 1, the encoding process is just the matrix multiplication described before. However, we should treat all of the entries as boolean values, since they represent only the presence of an odorant or an activation of a receptor. In implementation, these values are set to either 0 or 1, and the result of odorant response $\mathbf{R}$ may contain values larger than 1, but in condition statements, we only determine whether they are 0 or not.

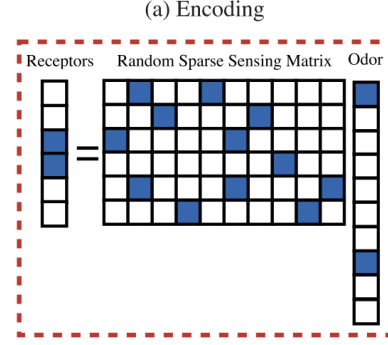


FIG. 1. Binary Encoding: The odor mixture and sensing matrix are represented as a binary vector and matrix. Rows of the sensing matrix represent sets of random different odorants. Each row or set corresponds to a one of the receptors in the receptor vector. If any of the odorants in the row are present in the odor mixture, then that receptor is considered present, as indicated by the solid filled box in the receptor vector in the figure. Otherwise, the receptor is considered inactive as indicated by the blank box in the receptor vector in the figure. As shown above, since odorants 1 and 8 are present in the odor mixture, the 3rd row of the sensing matrix includes the 1st odorant, and the 1st and 8th odorants appear in the sensing matrix only in the 3rd and 4th rows, respectively, only the 3rd and 4th receptors are considered active.

The decoding process is shown in Figure 2. We are treating a case with $N_L = 10$ and $N_R = 6$. We first identify the receptors that are not responding, which are R_1 , R_2 , R_5 , and R_6 . We first set all entries in the predicted odor vector $\mathbf{c}'$ to 1. Then, we look at the respective rows of the sparse sensing matrix, and identify the odors that are bound to these non-responding receptors. Next, we set the value of odors that we identified in the previous step to 0 in the vector $\mathbf{c}'$. After going over all of the non-responding receptors, we finalized our value of vector $\mathbf{c}'$. We compared it to the vector $\mathbf{c}$ to determine whether the decoding is correct.

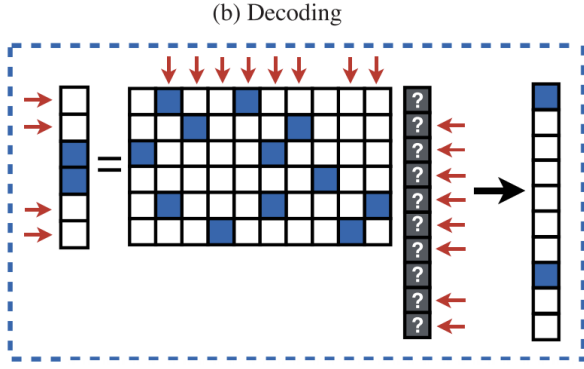


FIG. 2. Binary Decoding: Ignoring the original odor vector, the inactive receptors and corresponding sensing matrix rows from the encoding process are used to find each odorant that is determinedly not present in the odor mixture. Assuming all other odorants were present in the original odor, a guess for the odor mixture’s identity is created. As shown above, since receptors 1, 2, 5, and 6 are inactive and their corresponding rows in the sensitivity matrix contain odorants 2, 3, 4, 5, 6, 7, 9, and 10; these odorants are known to be absent in the odor mixture. Thus the guess for the odor includes only odors 1 and 8, which matches the original odor from figure 1.

We also tried various ways to optimize the execution efficiency of our code. We know that for `numpy` operations, it would be the fastest if we can call functions that are already implemented in `numpy`. We tried to eliminate all condition statement and loops as best as we could. Then, we used generative AI to rewrite our code using the `pytorch` library to utilize GPU acceleration. This speeds up our execution by approximately 5 to 10 times.

B. Weighted

For the weighted case, we focus on the existence of odorants along with their concentrations. Our mission is to find out under different values of N_R and K , how close the predicted odorant mixture $\mathbf{c}'$ is to the input odorant mixture $\mathbf{c}$.

As shown in Figure 3, the encoding process is just the matrix multiplication described before. This time, all entries in the odor vector $\mathbf{c}$, the sparse sensing matrix S , and the receptor response $\mathbf{R}$ are floating point numbers representing the concentration of the odorant, the strength of the binding between odorant and receptor, and the strength of odorant response.

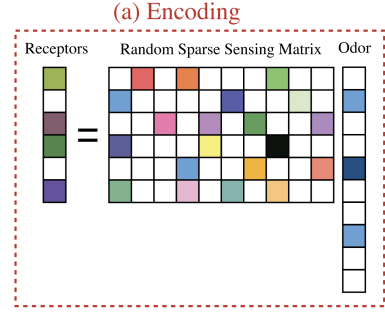


FIG. 3. Continuous Encoding: The odor mixture and sensing matrix are represented as a vector and matrix where present odorants are represented by a shaded box in which the color saturation indicates the level of concentration. Rows of the sensing matrix represent sets of random different odorants. Each row also corresponds to a one of the receptors in the receptor vector. If any of the odorants in the row are present in the odor mixture, then that receptor is considered present and has a bonding affinity, based on the concentration of that odorant in the mixture and indicated by the saturation of the color in the corresponding box. Otherwise, the receptor is considered inactive as indicated by the blank box in the receptor vector in the figure. In the example shown above, odorants 2, 5, and 8 are present in the odor mixture. Odorants 2 and 8 are then present in row 1, giving the first receptor a fairly strong bonding affinity. Likewise, rows 3, 4, and 6 also contain some combination of these odorants, which gives them fairly strong bonding affinities.

The decoding process is shown in Figure 4. In the same way, we are treating a case with $N_L = 10$ and $N_R = 6$. However, the process for the weighted case is slightly different. The elimination process is similar to the step described in the binary case. After doing the elimination, we conduct the concentration estimation, which is marked in the right-most of Figure 4. In this step, we are basically solving a linear equation with 4 variable and 4 equations. For actual cases, the number of equations should exceed the number of variables most of the time, since we have already eliminated so many odorants in the elimination process.

We weren’t able to achieve hardware acceleration in this simulation. This is because `pytorch` was not able to solve linear equations on the MacBook GPU.

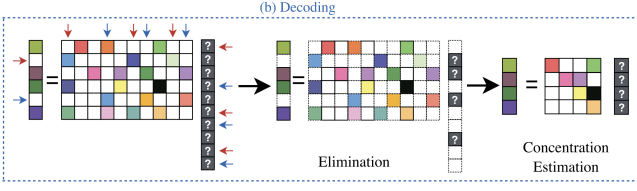


FIG. 4. Continuous Decoding: Ignoring the original odor vector, the inactive receptors and corresponding sensing matrix rows from the encoding process are used to find each odorant that is determinedly not present in the odor mixture. Then, each row in the sensing matrix corresponding to an inactive receptor and each column of the sensing matrix corresponding to an absent odorant is removed from the matrix, significantly reducing the shape of the matrix. The inactive parts of the receptor vector are then also removed and we use the simplified matrix to create a linear problem which we solve to get a guess for the odor mixture vector. In the example above, receptors 2 and 5 are inactive, meaning that odorants 1, 6, 9, 4, 7, and 10, which are in the 2nd and 5th rows of the sensing matrix, are not present in the odor. Thus, the 2nd and 5th rows are removed from the matrix and receptor vector and the 1st, 6th, 9th, 4th, 7th, and 10th columns are removed from the matrix. This leaves a linear equation with 4 unknowns and 4 equations, which can be solved numerically.

IV. RESULTS & DISCUSSION

A. Binary

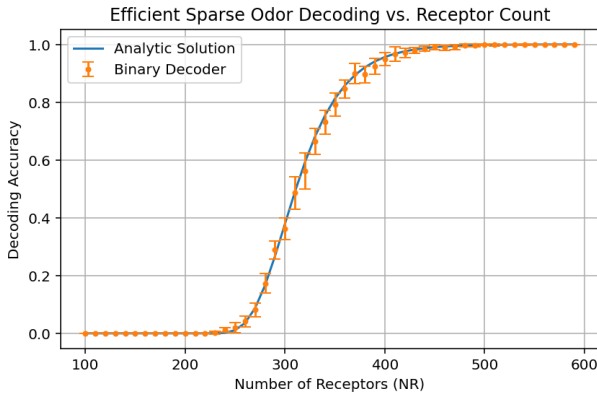


FIG. 5. Correct decoding probability as a function of the number of receptors given $N_L = 10000$, $s = 0.05$ and $K = 10$. Each data point represents the average correct decoding probability for 10 simulations each with 1000 trials and the error bar is ± 1 standard deviation across the 10 simulations. The line shown behind the points represents the analytical solution derived in the study and shown in equation 1.

From figure 5 we see that the decoding accuracy of the odor mixture for the case where odorants and receptor responses are binary becomes nonzero at around 250 for odor mixtures with an average of 10 odorants out of 10,000 possible odorants and a 5% chance for elements

of the sensing matrix to be nonzero. To put this in context, humans have around 300 nasal receptors, which, according to our plot, gives them around a 40% chance of successful odor identification using only information from the inactive receptors given the same initial conditions we used. On the lower end of this range, flies have an average of only 50 olfactory receptors, meaning that according to our model, if they were to use only the information from their inactive olfactory receptors, then they would have a 0% chance of correctly identifying odor mixtures with the complexity that we simulate. In contrast, animals with more sophisticated olfactory systems, such as dogs, cats, or rats, have upwards of 1000 olfactory receptors on average. According to our model, these animals have an average 100% chance of correctly identifying odors with similar complexity that we used, assuming that the behavior of the plot stays the same after converging.

The analytic solution plotted with the binary decoder points is based on a formula that was derived in the study. This derivation is beyond the scope of this paper but the formula is shown below in equation 1.

$$\left[\alpha + (1 - \alpha)(1 - [1 - s(1 - s\alpha)^{(N_L-1)}]^{N_R}) \right]^{N_L} \quad (1)$$

Where $\alpha = \frac{K}{N_L}$ is the probability that an odorant is present in the odor mixture.

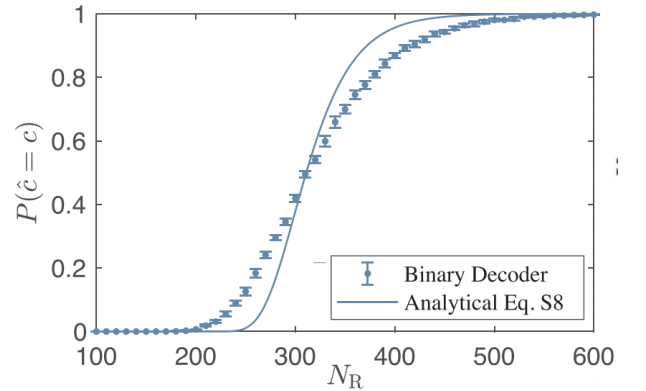


FIG. 6. The first plot we aim to reproduce from the study depicting the same as relationship we calculated in figure 5 for the binary odor mixture experiment. This graph also used 10 replicate simulations each with 1000 trials to find the mean decoding accuracy and error bars. The analytic solution is identical to that in figure 5.[1]

We can compare this relationship to the one found in the study, which is depicted in figure 6. While plots are overall very similar, two main differences stick out. Our plot is seemingly closer to the analytic solution than the plot from the study. Our plot, however, does have much more significant error bars, which is one possible reason for this discrepancy, but it seems unlikely that ours would have a better fit to the analytical solution if these values were not correct. On the other hand, this result

would raise the question of why the results of the study do not match the analytical solution, since the error bars on these points are so small. Since the paper does not directly state the algorithm used for their model, there is no direct way to compare our method to the one used in the study and, thus, no way to definitively answer this question of discrepancy.

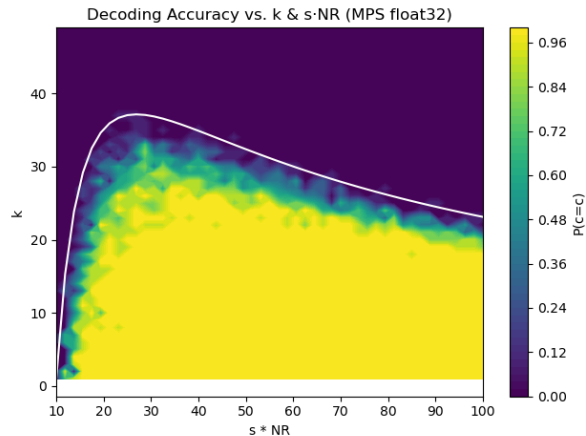


FIG. 7. Correct decoding probability as a function of the average number of odorants in each odor mixture (K) and the average number of receptors responding to odorants ($s * N_R$). For this simulation, N_R is held constant at 1000 and N_L is held constant at 10000. The average number of odorants in each odor mixture for this simulation range from 0 to 49 in integer steps of 1 and the average number of receptors responding to odorants ranges from 10 to 100 in 49 equal steps. This creates a 49 by 49 grid of data points. At each point, the probability is averaged over 10 trials before being plotted. The white line shown on the plot corresponds to $P(\hat{c} = c) = 0.5$.

In figure 7, we see the relationship between the accuracy of the decoding, the average number of odorants in the mixture, and the average number of receptors responding to an odorant. We find that the decoding accuracy ($P(\hat{c} = c)$) increases with the average number of receptors that respond to an odorant, reaching a success probability of nonzero probability for accurate odor identification starting around $s * N_R = 12$ for values of K that are less than 30. We can use these results to make similar contextual comparisons to the ones we made in the graph of decoding accuracy vs. number of receptors. Since this plot is created using a constant value for N_R , however, we can only draw reliable conclusions for animals with around 1000 receptors. If we assume that this distribution truly only depends on the product of $s * N_R$ and not either individually, however then we can estimate what this plot suggests for other animals as well. For flies, we see that for odor mixtures with an average number of odorants less than 30 out of 10000 possible, the probability that an odorant binds to a receptor would need to obey $s \geq 24\%$ in order for a fly to accurately predict the odor mixture. Similarly, for humans to have sim-

ilar chances of correctly producing an odor mixture of this level of complexity, the probability that an odorant binds to a receptor would need to obey $s \geq 5\%$, and for dogs, cats, or rats, it would need to obey $s \geq 1.5\%$.

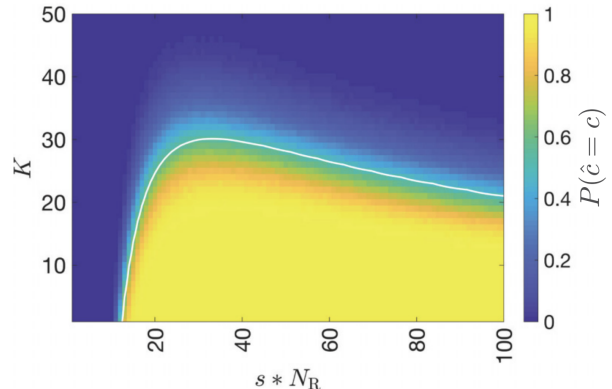


FIG. 8. The second plot we aim to reproduce from the study depicting the same as relationship we calculated in figure 7 for the binary odor mixture experiment. For this simulation, $N_L = 10000$ and $N_R = 1000$, as they did for our simulation. It is unclear exactly how many different values of K and s are used, but the decoding accuracy at each point on the grid is averaged over 10000 trials. The white line corresponds to $P(\hat{c} = c) = 0.5$. [1]

As we did for the first graph, we can compare this three-dimensional relationship to the one that we aim to reproduce from the study, shown in figure 8. The first clear distinction between the two is that our graph is much noisier. For our simulation, we used 49 steps for both K and s , giving us a grid of 49 by 49 points in our plot. Although the research paper we are aiming to reproduce does not explicitly state how many different data points were used for their plot, the evenness and lack of noise in the plot suggests a much denser grid of points than ours. Furthermore, for run-time reasons, we were only able to average the probability of the odor mixture being identified correctly over 10 trials, whereas it is noted in the research paper that they used 10000 trials per data point to average this value. Aside from this discrepancy in noise, the only other notable difference between our plot and the results of the paper is the relative shift of the high probability region to the left in our graph. This is simply the result of our $s * N_R$ values starting at 10 rather than 0 to avoid any errors that might arise from a 0 value for $s * N_R$. Overall, both of our plots for the binary case qualitatively and quantitatively support the results from the plots in the paper by Singh, Tchernookov, and Balasubramanian. [1]

B. Weighted

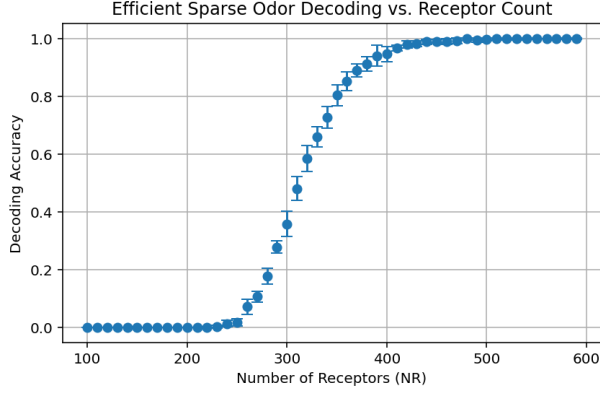


FIG. 9. Correct decoding probability as a function of the number of receptors for the case where each odorant, as well as receptor affinity, is represented by a weighted value between 0 and 1. Each data point represents the average correct decoding probability for 10 simulations each with 1000 trials and the error bar is ± 1 standard deviation across the 10 simulations.

Looking at figure 9, we can see that in our model the relationship between the number of receptors and the decoding accuracy for the continuous decoder is very similar to that of the binary decoder. In fact, by direct comparison, we find that these two graphs are almost identical. The accuracy of the decoder starts to become nonzero around $N_R = 250$, as it did before, and converges to 1 shortly after $N_R = 400$ for an odor mixture with 10 nonzero odorants presences out of 10000 possible odorants and a 5% chance of an odorant bonding to a receptor. This would suggest that having weighted odorants and bonding affinity does not have any considerable effect on the accuracy of the decoding process. However, this is not what was found in the research paper.

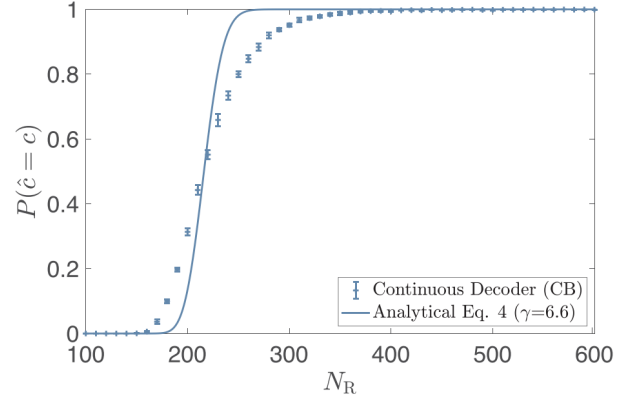


FIG. 10. The third plot we aim to reproduce from the study depicting the same as relationship we calculated in figure 9 for the weighted odor mixture experiment. This graph also used 10 replicate simulations each with 1000 trials to find the mean decoding accuracy and error bars. The line plotted behind the points from the continuous decoder is an analytic solution based on a normal cumulative distribution function that was derived in the research paper.[1]

The results from the paper for the relationship we found in figure 9 are shown in figure 10. As can be seen in the figure, the probability of correctly identifying the odor mixture using the continuous decoder becomes nonzero starting at N_R values around 160 for an odor mixture with 10000 possible odorants, 10 of which have a nonzero concentration and a 5% chance for an odorant to bind to a receptor. This would suggest that considering odorant concentration and receptor bonding affinity actually improves the results of the decoding algorithm. This seems counter-intuitive, but since the algorithm is not exactly the same as the binary case, it is not impossible. This earlier decoding success does not match the results we obtained, since our result has nearly the same relationship as the binary case. As described in our methods section, the decoding process for our continuous decoder is fairly similar to that of the binary case. Aside from removing the rows and columns corresponding to the inactive and absent elements from the sensing matrix and solving a linear equation to find the original odor concentrations, the elimination process based on the inactive receptors is not significantly different from the one used in the binary decoder. As a result, the only complication that should make our results differ from the binary decoder would be if the linear equation were not solved. This would mean that our model could not be more accurate than the binary decoder regardless of the number of receptors. As in the binary case, the exact algorithm used in the paper for the continuous decoder is not explicitly stated, so our algorithm is based on as much information about the algorithm as we could implement from the paper. The only aspect of the algorithm mentioned in the paper that is not implemented in our model is treating receptors that respond below a certain threshold as being inactive. This process is only briefly mentioned, and

no threshold is explicitly given, so we did not implement this in our algorithm. Treating more receptors as inactive would further reduce the linear problem that we solve to make the guess for the odor mixture in the decoder, so, in theory, this could improve the accuracy of the continuous decoder for fewer receptors. As a result, this is the most likely source of the discrepancy between our results and the results found in the paper for the continuous decoder.

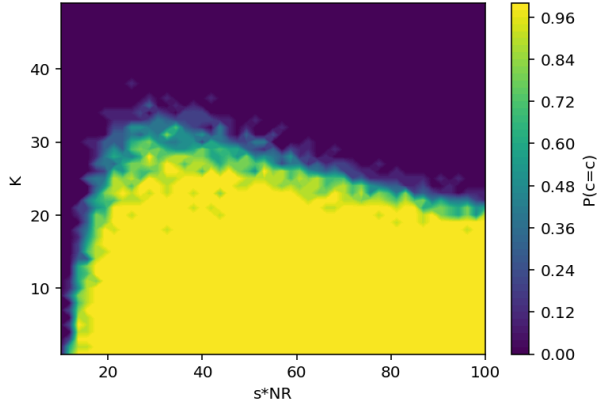


FIG. 11. Correct decoding probability as a function of the average number of odorants in each odor mixture (K) and the average number of receptors responding to odorants ($s * N_R$) for the case where each odorant, as well as receptor affinity, is represented by a weighted value between 0 and 1. For this simulation, N_R is held constant at 1000 and N_L is held constant at 10000. The average number of odorants in each odor mixture for this simulation range from 0 to 49 in integer steps of 1 and the average number of receptors responding to odorants ranges from 10 to 100 in 49 equal steps. This creates a 49 by 49 grid of data points. At each point, the probability is averaged over 10 trials before being plotted.

The three-dimensional relationship that we calculated for the binary case in figure 7 is shown for the weighted case in figure 11. As with our other result for the continuous decoder, this plot looks very similar to the one we found for the binary decoder. Due to the same reasons previously described, this makes sense for our model, but once again does not agree exactly with the results from the paper. As shown in figure 12, in the paper, they found that the probability of accurate decoding quickly becomes nonzero at around $s * N_R = 8$ for K values less than 40. This means that the continuous decoder in the study done in the paper is more accurate at predicting the odor mixture when $s * N_R$ is lower than the binary decoder is and can also accurately predict a more complex odor mixture. The probability curve then flattens out and begins to match that of the binary decoder and our results. This further suggests that the discrepancy between our results and the results from the paper is primarily at lower numbers of receptors and more complex odor mixtures. Overall, we can hypothesize that our results differ due to the elimination of low-response receptors, as described before.

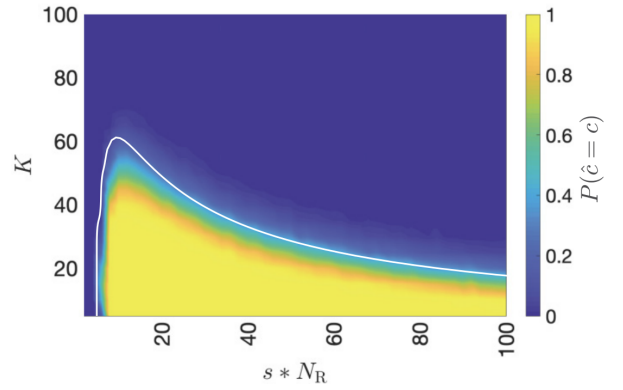


FIG. 12. The second plot we aim to reproduce from the study depicting the same as relationship we calculated in figure 11 for the weighted odor mixture experiment. For this simulation, $N_L = 10000$ and $N_R = 1000$, as they did for our simulation. It is unclear exactly how many different values of K and s are used, but the decoding accuracy at each point on the grid is averaged over 10000 trials. The white line corresponds to $P(\hat{c} = c) = 0.5$. [1]

V. CONCLUSION

Our goal for this computational model is to reproduce the results of a study on odor identification by elimination by Vijay Singh, Martin Tchernookov, and Vijay Balasubramanian [1]. Specifically, we aim to answer the question of how useful inactive olfactory receptors are in identifying odor mixtures. Since olfactory receptors bind to a wide range of odorants, an inactive receptor indicates that none of the odorants that bind to that receptor are present in the odor mixture, meaning that a set of inactive receptors can reveal many odorants that are not in the odor mixture, significantly reducing the different odors that the mixture could represent. Most natural odors are made up of 10-40 odorants out of 10000 total possible odorants. Our results support the results found in the aforementioned study, which suggest that odors of such complexity can be encoded in, and decoded from, a network of relatively few receptor types which each bind to roughly 5% of odorants [1]. The results further suggest that a greater number of receptors leads to a more reliable prediction of odors for more complex odors. However, since we are specifically looking at odor identification using inactive receptors, more research would be necessary to draw conclusions about the overall relationship between complex odor identification and the number of receptors.

Both our model and the one from the study use the assumption of steady-state dynamics, which create a much simpler model, but do not account for physical phenomena such as sniffing or temporary exposure to an odor mixture. Thus the results of our model are slightly less applicable to a realistic olfactory system, but more research could be done using our model as a basis to ex-

plore the highly dynamic process of the olfactory system to make it more applicable. Having generally supported the results of the study, our results could also be consid-

ered in this research and could also be applied to research in artificial odor identification, such as robotic odor detection.

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- [1] V. Singh, M. Tchernookov, and V. Balasubramanian, What the odor is not: Estimation by elimination, *Physical Review E* **104**, 024415 (2021).